

An Experimental Study of CNN-Based Traffic Sign Recognition for Autonomous Driving Scenarios

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- ☒ ~~Track 1 - Research~~
- ☐ Track 2 - Application
- ☐ Track 3- Blue Sky Idea

Link to Implementation: [🔗 CAP4630 Final Project - Jonathan Yunes.ipynb](#)

Abstract

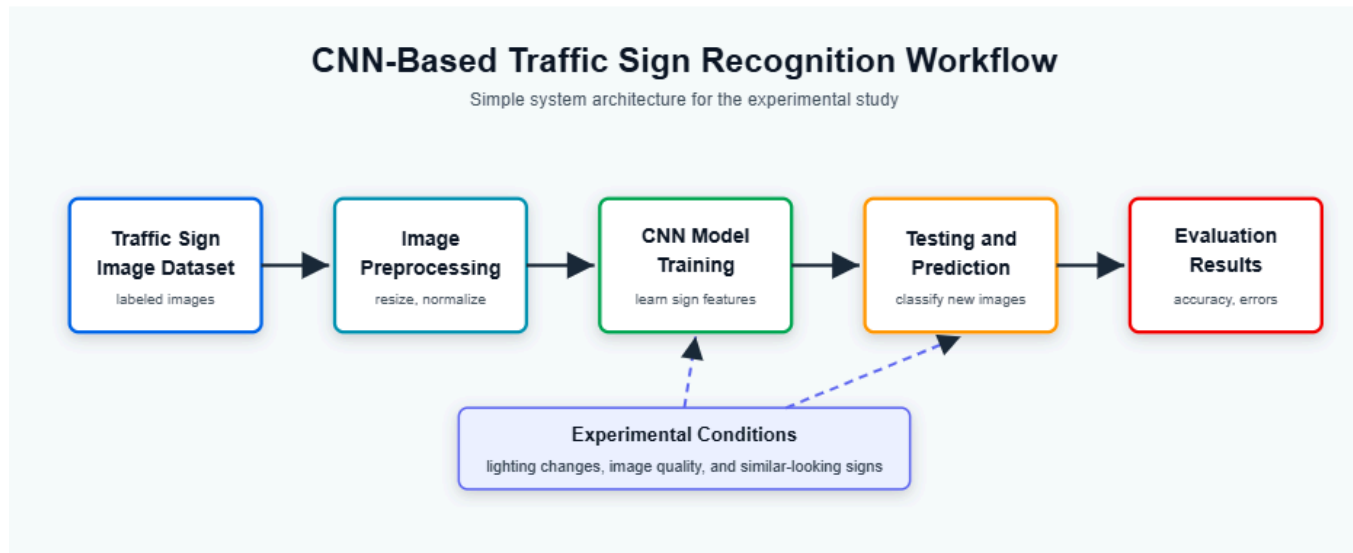
This project investigates the problem of recognizing traffic signs from images in the context of Artificial Intelligence. Traffic sign classification and recognition is a crucial task for autonomous driving systems. Existing approaches to this problem often face challenges such as changes in lighting, image quality, and similarities between certain traffic signs. In this study, I’m going to use a convolutional neural network (CNN) to analyze how well deep learning models can perform traffic sign recognition. To study this idea, I’m going to conduct some experiments by training and testing the CNN on a simplified version of the LISA (Laboratory for Intelligent and Safe Automobiles) Traffic Sign S1gn dataset, which was downloaded from Kaggle, and evaluate its performance under different conditions.

Introduction

The key question guiding this project is how well a convolutional neural network can recognize and classify traffic signs from images, which is important for autonomous driving systems. This could be challenging due to variations in lighting, quality of the image, and the similarity between certain signs. These challenges can make it harder for the models to correctly classify the signs in real-world scenarios.

To explore this problem, I’m creating a small experimental study using the LISA traffic sign dataset and a CNN-based model. The goal of this project is to evaluate how the model performs and to learn about the strengths and weaknesses. I believe that the CNN model can achieve a strong performance. However, the accuracy might be affected by the conditions of the image.

Proposal



In this project, I propose a system which will use a CNN to address the problem of traffic sign recognition. The main goal of this approach is to allow the CNN to automatically learn the visual features from the images like shapes and colors so it can classify the different traffic signs. The diagram above shows the workflow for my solution. The main components will include the dataset, preprocessing techniques, the actual CNN model, and the evaluation metrics. The design should allow the system to process the input data and produce predictions. It'll do this by passing the images through the trained CNN model. The output will be the predicted traffic sign based on the patterns learned during the training of the model.

Experiments

Experiment 1 - Baseline CNN Model

In this experiment I tested how well a basic CNN model performs on traffic sign recognition. The model was trained using the LISA dataset with preprocessing techniques like resizing and normalizing the images. This is the starting point and this experiment shows us how the model performs on its own.

The model got an accuracy of 87.22% and a loss of 0.3413. These results tell us that this baseline CNN model performed fairly well. We're going to compare these results to the other experiments.

Experiment 2 - CNN with Data Augmentation

For this experiment, I tested whether applying data augmentation would improve the model's performance. I applied techniques like rotation, flipping, and brightness to increase the variation of the images. This helps simulate real-world conditions where traffic signs may appear in different orientations.

I used the same baseline CNN architecture so the results were fair. The model got an accuracy of 52.22 and a loss of 1.2255. As a result, compared to the baseline model, the model's performance declined. This tells us that data augmentation made the classification more difficult for the model. Also, this means that the architecture isn't complex enough to handle the enhanced data effectively.

Experiment 3 - CNN Under Challenging Conditions

I tested how CNN performs when the images are more difficult to recognize. I modified the dataset to simulate challenging conditions like low brightness and slight zoom changes. These changes represent situations where the signs may not always be clearly visible.

Again, the same architecture for the model was used. After the training, the model got an accuracy of 81.11% and a loss 0.5032. So compared to Experiment 2, the performance improved. This means that the model could handle challenging conditions better than data augmentation.

Conclusion

This project explored how a CNN performs on traffic sign recognition under different conditions. The baseline model showed the highest accuracy, telling us that the model worked well on standard data. The performance decreased when data augmentation was applied. On the other hand, it performed better under moderate challenge conditions. Overall, the results show that while CNNs are effective for traffic sign recognition, the quality or type of data could have an impact on the performance of the model.